Application of Remote Sensing for Long-Term Analysis of Environmental Criticality Indices in Kendari City

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ABSTRACT

The decline in vegetation cover has a crucial impact on ecological quality and function, as seen on critical land. Kendari City experiences fairly rapid population growth, being the capital of Southeast Sulawesi Province. This study analyzes long-term changes in Kendari City's environmental quality using the Environmental Criticality Index (ECI) derived from Multitemporal Landsat Satellite Imagery. Normalized Difference Vegetation Index (NDVI) and Land Surface Temperature (LST) can provide information on environmental criticality through the contrast difference between vegetation as a cooling agent and surface temperature, which is often amplified by built-up areas. The results of the study show that the concentration of ECI occurs in the city center area that stretches along Kendari Bay. The moderate ECI experienced regional fluctuations in the 1991-2002, 2001-2014, and 2014-2023 periods, namely -8.43%, +2.28%, and -20.35%, respectively. Meanwhile, the high class experienced gradual expansion, namely +3.38%, +5.12%, and +1.03%, respectively. The 2001-2014 period was the peak phase of ECI changes, with a tendency for environmental quality to decline. The Urban-Rural Gradient shows that the highest ECI value occurs in urban areas and decreases as it moves towards the suburbs, where there are still rural areas. The correlation test shows a strong relationship between them and indicates a significant influence on ECI. These findings underscore the importance of sustainable reforestation strategies in reducing urban environmental pressures.

Keywords: Land Surface Temperature (LST), Normalized Difference Vegetation Index (NDVI), Environmental Criticality Index (ECI), Urban-Rural Gradient, Environmental Degradation.

1. Introduction

Cities play a crucial role in supporting human life by serving as centers of housing, government, economy, and social interaction. However, rapid urbanization often brings complex environmental challenges. Population growth increases demand for land, driving massive conversion of natural to built-up areas, disrupting the balance of urban ecosystems, and affecting air quality, water availability, surface temperature, and vegetation cover. [1]. In this context, vegetation plays a key role in maintaining thermal comfort and environmental stability in urban areas.

Kendari City, the capital of Southeast Sulawesi Province, is also experiencing this dynamic. The region's population density continues to increase, reaching 355,665 people as of June 30, 2024 [2], placing significant pressure on its land resources. The Urban Heat Island (UHI) phenomenon in Kendari has been identified as expanding by up to 18.92% of its total area [3]. Like other rapidly developing provincial capitals, the city still has substantial potential land for future development. However, further expansion could lead to environmental degradation and disruption of the ecological balance [4], [5]. These changes ultimately increase urban environmental criticality, defined as the degree of imbalance between the physical and ecological components of the environment.

Efforts to overcome these challenges require accurate monitoring tools, spatially comprehensive and cost-effective. Remote sensing technology offers these advantages by enabling continuous observation of land surface conditions across space and time, and facilitating multitemporal analysis of urban growth, sprawl, and environmental change [6], [7]. Halefom et al. [1] emphasize that integrating remote sensing with geoinformation technologies, such as GIS, spatial analysis, and dynamic modelling, is crucial for monitoring and understanding urban environmental dynamics.

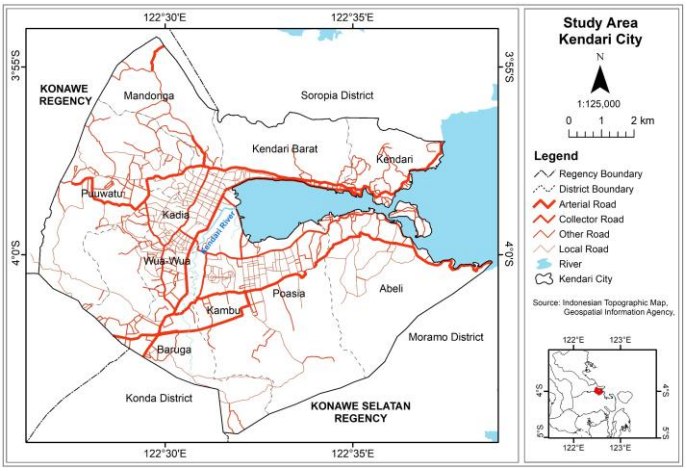


Figure 1. Study Area of Kendari City

Among various satellite datasets, Landsat imagery is one of the most widely used sources for this purpose [8], [9]. Landsat imagery provides reliable data through the Environmental Criticality Index (ECI). The ECI index is an index that integrates vegetation and thermal indicators towards environmental degradation and urban sustainability [5], [10]. Higher ECI values indicate greater environmental stress and decreased ecological stability. The Normalized Difference Vegetation Index (NDVI) is commonly used to represent vegetation density, producing robust and consistent results across a wide range of environmental conditions [11]. Meanwhile, the Land Surface Temperature (LST) algorithm identifies variations in surface heat that reflect land cover conditions. The interaction between NDVI and LST forms the basis for deriving ECI values that provide a comprehensive indicator of urban environmental stress [5], [10], [12].

The use of NDVI- and LST-based ECI is grounded in theoretical and empirical considerations that vegetation cover and surface temperature are the most direct and fundamental determinants of environmental criticality and are broadly applicable across different biophysical contexts [12], [13], [14]. In contrast, ECI studies that incorporate numerous additional indices without prior elimination risk multicollinearity and bias due to overlapping spectral sensitivities, particularly in heterogeneous landscapes, which can obscure the fundamental relationship between the core indices [15], [16].

Previous studies have demonstrated that areas with reduced vegetation cover and higher building densities tend to experience increased surface temperatures and environmental criticality [14], [17], [18], [19]. However, most existing studies have focused on low-lying urban areas with limited spatial coverage and short-term observation periods, limiting insights into environmental criticality dynamics in more complex urban landscapes [20], [21], [22].

Kendari City presents a distinct case, as it is located around Kendari Bay and is surrounded by hilly terrain with varying slopes. Although coastal hilly areas are typically cooler, the influence of urbanization, air pollution, and the UHI effect can alter the local microclimate, resulting in unpredictable thermal patterns [23]. In addition, few studies have employed a multitemporal approach that simultaneously integrates NDVI, LST, and ECI to capture long-term environmental dynamics across an urban–rural gradient, particularly in a tropical coastal city like Kendari City.

Based on the description above, this study aims to (1) analyze the long-term trend of environmental criticality in Kendari City using NDVI and LST indices, and (2) examine the spatial variation of environmental criticality from the city center to the suburbs. This study contributes to a more comprehensive understanding of how land use change and urban growth affect the sustainability and resilience of tropical coastal-hilly cities.

2. Methods

Study Area. The research location is in Kendari City, Southeast Sulawesi Province. Geographically, this area is between 3º54'30" - 4º3'11" S and 122º23' - 122º39'E, covering a total area of 22,646.37 ha (Figure 1). The land cover types in this region include several main categories, such as built-up areas, rice fields, vegetation, shrubs, dry fields, water bodies, and open land. In addition, there are specific land cover types such as forests, mixed gardens, mangroves, and fishponds. Kendari City is divided into 10 Districts, namely Abeli, Baruga, Kadia, Kambu, Kendari, West Kendari, Mandonga, Poasia, Puuwatu, and Wua-Wua. Kendari is well-suited for long-term environmental vulnerability analysis as it is a rapidly growing coastal city with fragile ecosystems and a history of economic activities that have influenced its environmental conditions. These factors make Kendari a relevant case study for

understanding the environmental impacts of urban development and designing strategies for sustainability.

Data Source. This study uses a multitemporal Landsat satellite imagery dataset from 1991 to 2023. Landsat 5 TM data for 1991. Landsat 7 ETM+ for 2001, and Landsat 8 OLI/TIRS for 2014 and 2023. All images were obtained as Collection 2 Level-1 products, which have been orthorectified using ground control points (GCPs) and a digital elevation model (DEM) to ensure geometric accuracy and spatial consistency among different acquisition dates. All images share a spatial resolution of 30 meters, so no additional resampling was required. Each image was reprojected to the UTM Zone 51S - WGS84 coordinate system to ensure alignment across all years. The entire Landsat satellite collection was downloaded from the United States Geological Survey (USGS) website with the path/row archive 112/063. The specifications of the data used are shown in Table 1.

Pre-processing. Since Level-1 data are not atmospherically corrected, radiometric and atmospheric corrections were applied manually. The digital number (DN) values were first converted into Top of Atmosphere (TOA) reflectance using metadata parameters (gain, bias, and solar elevation angle). Subsequently, the Dark Object Subtraction (DOS) method was employed to minimize atmospheric scattering and haze effects, resulting in surface reflectance values suitable for temporal comparison.

Cloud and shadow masking was performed using the Quality Assessment (QA) band, complemented by visual inspection to verify that only cloud-free pixels (less than 5% cloud cover) within the area of interest were used in subsequent analyses. All these steps are done with ArcMap 10.4.1.

Index Extraction from Landsat Collection. Extraction of Land Surface Temperature values on Landsat 5 and Landsat 7 uses Band 6, while on Landsat 8, it uses Band 10 and Band 11.

Landsat 5 TM and Landsat 7 ETM+. Quantitative information from Landsat 5 TM and Landsat 7 ETM+ images can be obtained by converting Digital Number (DN) to Physical Quantities (PQ), Radiance, and Brightness Temperature (BT). This is done to correct the image that was previously calibrated by atmospheric effects. Landsat 5 and Landsat 7 ETM+ images can be converted to Top of Atmosphere (TOA) radiance using the following Equation 1 [25]:

$$L_\lambda = \left(\frac{L_{MAX_\lambda} - L_{MIN_\lambda}}{Q_{cal_{MAX}} - Q_{cal_{MIN}}}\right)(Q_{cal} - Q_{cal_{MIN}}) + L_{MIN_\lambda} \quad (1)$$

where Q_{cal} is calibrated pixel values in quantization DNs, L_{MIN_λ} is the spectral emission corresponding to

Table 1. Specifications of Landsat 5 TM, Landsat 7 ETM+, and Landsat 8 OLI images used

Satelite Imagery	Band	Spatial Resolution	Path/Row	Date	Criteria
Landsat 5 TM	Band 1 – Blue	30	112/063	22 July 1991	AOI (Cloud cover < 5%)
	Band 2 – Green	30			
	Band 3 – Red	30			
	Band 4 – Near Infrared (NIR)	30			
	Band 5 – Shortwave Infrared (SWIR) 1	30			
	Band 6 – Thermal	120			
	Band 7 – Shortwave Infrared (SWIR) 2	30			
Landsat 7 ETM+	Band 1 – Blue	30	112/063	06 May 2001	
	Band 2 – Green	30			
	Band 3 – Red	30			
	Band 4 – Near Infrared (NIR)	30			
	Band 5 – Shortwave Infrared (SWIR) 1	30			
	Band 6 – Thermal Infrared	60			
	Band 7 – Shortwave Infrared (SWIR) 2	30			
Landsat 8 OLI/TIRS	Band 2 – Blue	30	114/064	09 October 2014	
	Band 3 – Green	30		09 April 2023	
	Band 4 – Red	30			
	Band 5 – Near Infrared (NIR)	30			
	Band 6 – Shortwave Infrared (SWIR) 1	30			
	Band 7 – Shortwave Infrared (SWIR) 2	30			
	Band 10 – Thermal Infrared (TIRS) 1	100			
	Band 11 – Thermal Infrared (TIRS) 2	100			

Source: [24].

$Q_{cal\ MIN}$ (Watts/(m^2.sr.μm)), L_{MAX_λ} is the spectral emission corresponding to $Q_{cal\ MAX}$ (Watts/(m^2.sr. μm)), $Q_{cal_{MIN}}$ is the minimum quantization calibrated pixel value corresponding to L_{MIN_λ} (DN=1), $Q_{cal_{MAX}}$ is the maximum quantization calibrated pixel value corresponding to L_{MAX_λ} (DN=255).

The Radiance value is then converted into Kelvin (K) using Equation 2 and then converted again into Celsius (C) units using Equation 3. The spectral radiance calibration constants 1 and 2 for Landsat 5 TM are = 607.76 and = 1260.56, while those for Landsat 7 ETM+ are = 666.09 and = 1282.71. Extraction of the Radiance value using the following Equation 2 [19], [26]:

$$T = \frac{K2}{ln\left(\frac{K1}{L_\lambda} + 1\right)} \quad (2)$$

where T is temperature (kelvin), L_λ is radiance (Watts/(m^2.sr. μm)), $K1$ is spectral radian calibration constant 1, and $K2$ is the spectral radian calibration constant 2.

$$BT = T - 273{,}15 \quad (3)$$

where BT is brightness temperature, and T is temperature in Kelvin

Landsat 8 OLI/TIRS. The quantitative extraction of Landsat 8 OLI/TIRS imagery follows similar stages to those of Landsat 5 TM. The TOA value is determined using Equation 4 [19], [23], while the Radiance value is determined using Equation 2. The Calibration Constants of spectral radiance 1 and 2 are different for Landsat 8 OLI/TIRS. The value of $K1$ = 774.8853, while the value of $K2$ = 1321.0789.

$$L_\lambda = 0{,}0003342 \times Q_{cal} + 0.1 \quad (4)$$

where L_λ is radian spectral from sensor (Watts/(m^2.sr.μm)), and Q_{cal} is calibrated pixel values in quantization DNs

The BT value is determined using the following Equation 5 [19], [23]:

$$BT = \frac{BT_{10} + BT_{11}}{2} \quad (5)$$

where BT is brightness temperature, BT_{10} is thermal infrared 1 band, and BT_{11} is thermal infrared 2 band.

Before converting BT to LST, it is important to calculate the land surface emissivity (e) value. Ahmad et al. [26] proposed a methodology using the Normalized Difference Vegetation Index (NDVI) estimation algorithm. Band 4 and Band 3 are used to calculate the NDVI estimate with the following Equation 6 [26]:

$$NDVI = \frac{(NIR - Red)}{(NIR + Red)} \quad (6)$$

where $NDVI$ is the vegetation density index value, NIR is band 4, and Red is band 3

After the NDVI value is obtained, the next step is to calculate the Vegetation Proportion (Pv) value using the following Equation 7:

$$Pv = \frac{(NDVI - NDVI_{Min})}{(NDVI_{Max} - NDVI_{Min})} \quad (7)$$

where Pv is the proportion of vegetation, $NDVI$ is the vegetation density index value, $NDVI_{Min}$ is the minimum NDVI value, $NDVI_{Max}$ is the maximum NDVI value.

After obtaining the Pv value, the (e) value is then calculated using the following Equation 8 [19], [23]:

$$e = m\,Pv + n \quad (8)$$

where e is land surface emissivity, m has a value of 0.004, Pv is the proportion of vegetation, n has a value of 0.986.

The LST estimate was calculated using Equation 9, and the temperature obtained in Kelvin units was converted to the Celsius scale using the following Equation [27]:

$$LST = \frac{BT}{\left[1 + \left(\frac{\lambda \times BT}{a}\right) ln\,(e)\right]} \quad (9)$$

where LST is land surface temperature, BT is the brightness temperature of TOA (°K), λ is radiance emission wavelength (Band 10=10.8; Band 11=12), a is the constant obtained from h×c/σ (1.4388 µmK), e is land surface emissivity.

The value of a is obtained from a combination of h, c, and σ. h is Plank's constant= 6.626×10^{-34}Js; c is the speed of light = 2.998×10^{8} ms^{-1}; and σ is Boltzmann's constant = 1.38×10^{-23} J K^{-1}.

Extraction of Environmental Criticality Index. The ECI is a composite index designed to represent the degree of environmental stress and degradation resulting from the interaction between vegetation condition and surface thermal characteristics. Conceptually, ECI reflects the imbalance between ecological components (represented by vegetation cover) and physical stressors (represented by land surface temperature), which together determine the level of environmental criticality in urban landscapes.

In urban environments, vegetation density plays a crucial role in regulating the microclimate, reducing surface temperatures, and maintaining ecological stability, whereas elevated land surface temperatures are commonly associated with urban expansion and anthropogenic heat emissions. Therefore, the integration of NDVI and LST provides a direct measure of environmental criticality, as areas with sparse vegetation and higher surface temperatures tend to experience greater environmental stress.

Before index calculation, NDVI and LST values are normalized using a linear stretch to a common scale of 1 to 255. This normalization serves two main purposes. First, it equalizes the spectral ranges of both indicators to ensure comparability and proportional contribution in the composite index. Second, excluding zero values prevents null pixel effects during raster multiplication, which could otherwise result in the loss of valid environmental information. The use of an integer-based 1–255 scale is also consistent with common practices in remote sensing–based index construction, facilitating computational stability and spatial analysis. The ECI calculation is done using the following Equation 10:

$$ECI = LST \times NDVI\ (Strech\ 1 - 255) \qquad (10)$$

where ECI is Environmental Criticality Index, LST is Land Surface Temperature value, and NDVI is Normalized Difference Vegetation Index.

The multiplicative formulation is employed to capture the interaction between the two factors, rather than treating them as independent additive components. This approach emphasizes areas where high surface temperature coincides with reduced vegetation cover, which are considered the most environmentally critical zones.

Analytical Techniques. A temporal analysis was conducted to examine changes in the spatial extent and trends of NDVI, LST, and ECI from 1991 to 2023. In this study, NDVI values were classified into four classes, namely non-vegetation (<0), low vegetation (0-0.2), moderate vegetation (0.21-0.5), and high vegetation (0.51-1) [22]. The LST is classified into three classes, namely low (< 22.1 °C), moderate (22.2 °C - 25.1 °C), and high (> 25.2 °C) [28]. Meanwhile, ECI is classified using the natural breaks/Jenks method into low, moderate, and high classes [20]. The area of each class for every period was calculated to identify long-term patterns in the formation of the criticality index and its outcomes.

To explore the internal relationships among these indices, a pixel-based correlation analysis was conducted by extracting raster values at corresponding spatial locations (n=1301). The relationships between NDVI–ECI and LST–ECI were examined descriptively to illustrate how variations in vegetation greenness and surface temperature are reflected in the composite ECI values (as measured by the coefficient of determination, R^2). It is important to note that this analysis is not intended as an independent statistical validation, as NDVI and LST are mathematical components of the ECI formulation. Accordingly, the correlation results are interpreted as an assessment of internal consistency and spatial coherence rather than as evidence of independent explanatory power [29], [30].

In addition, an urban–rural gradient analysis was applied by drawing a linear transect extending from the city center toward suburban and rural zones. The transect lines were then converted into sample points (n=167) with 30 m intervals. NDVI, LST, and ECI values along this transect were used to observe spatial variations and the intensity of urban influence on the basic indices forming the ECI. This approach enables visualization of gradual environmental transitions along the urbanization continuum and helps contextualize the spatial behavior of the ECI across different urban intensity levels. This method has been widely applied in urban ecological and thermal environment studies [31], [32].

3. Result and Discussion

Normalized Difference Vegetation Index Trends. The NDVI algorithm is an index used to identify the distribution of vegetation density levels. The greener the color, the higher the vegetation density.

High vegetation density is generally distributed in the suburbs, while low vegetation density is found in the city center (Figure 2). It is known that the low vegetation density class increased to 2,613.06 ha in 2023 (Figure 3). This figure represents a significant increase in commercial and built-up areas, threefold compared to 1991. In the medium density class, the area decreased to 5,433.93 ha. A similar trend was also demonstrated by classes with higher vegetation density, where approximately 7,364.34 ha of forest were lost over 32 years. However, the most severe period was from 1991 to 2001, when approximately 9,144.54 ha were lost.

According to Indriyani et al. [21], NDVI can be used to represent vegetation presence. Areas with high vegetation density generally represent green open spaces, residential parks, and shade trees on streets. Meanwhile, the further away from the city outskirts, the higher vegetation density is more representative of forest cover, which contains woody vegetation with very dense leaves. Areas with lower density indicate sparse or even no vegetation, while denser land cover indicates forest areas [33].

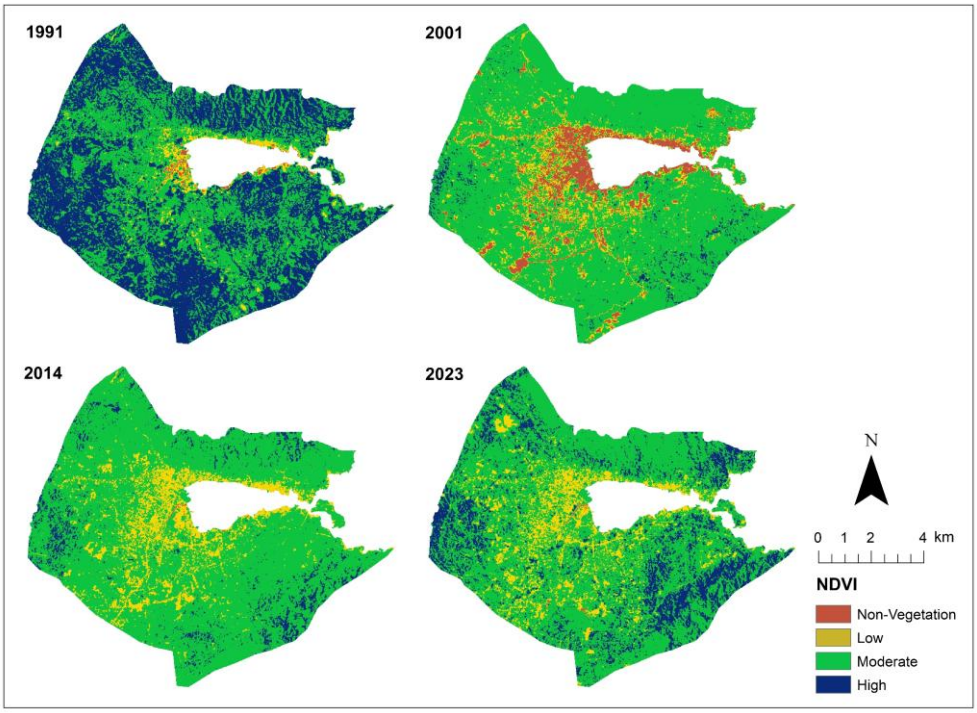
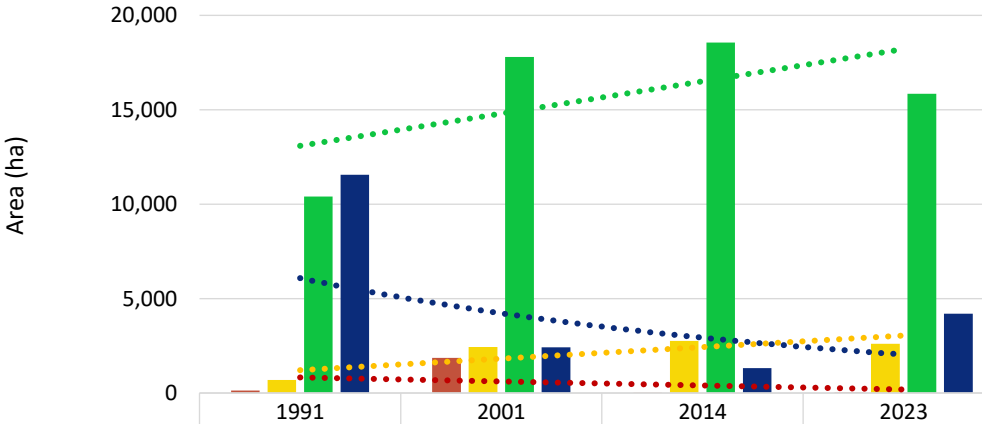


Figure 2. Distribution of Vegetation Density Index (NDVI) in Kendari City 1991-2023

Non-vegetation and high vegetation areas showed a decrease of 0.43% and 32.52%, respectively, while low and moderate vegetation areas experienced an increase of 8.52% and 23.99%, respectively, from 1991 to 2023. These changes can be attributed to the phenomenon of urbanization and population growth. This is consistent with other image monitoring studies that the rate of urbanization, infrastructure development, land conversion, and the filling of water bodies contribute to these changes [34], [35]. Kendari's position has attracted residents from surrounding areas to migrate to the city, thereby increasing socio-economic pressures that have led to the expansion of built-up areas and the conversion of non-built-up land. This situation also occurs in other cities in Indonesia, especially in relatively smaller areas [36].

Land Surface Temperature Dynamics. Land Surface Temperature is defined as the temperature



	1991	2001	2014	2023
Non-Vegetation	126.36	1,852.65	17.10	28.71
Low	682.65	2,442.15	2,766.06	2,613.06
Moderate	10,407.69	17,792.73	18,567.63	15,841.62
High	11,556.03	2,411.49	1,312.68	4,191.69

Figure 3. Distribution and Trend of NDVI Area in Kendari City 1991-2023

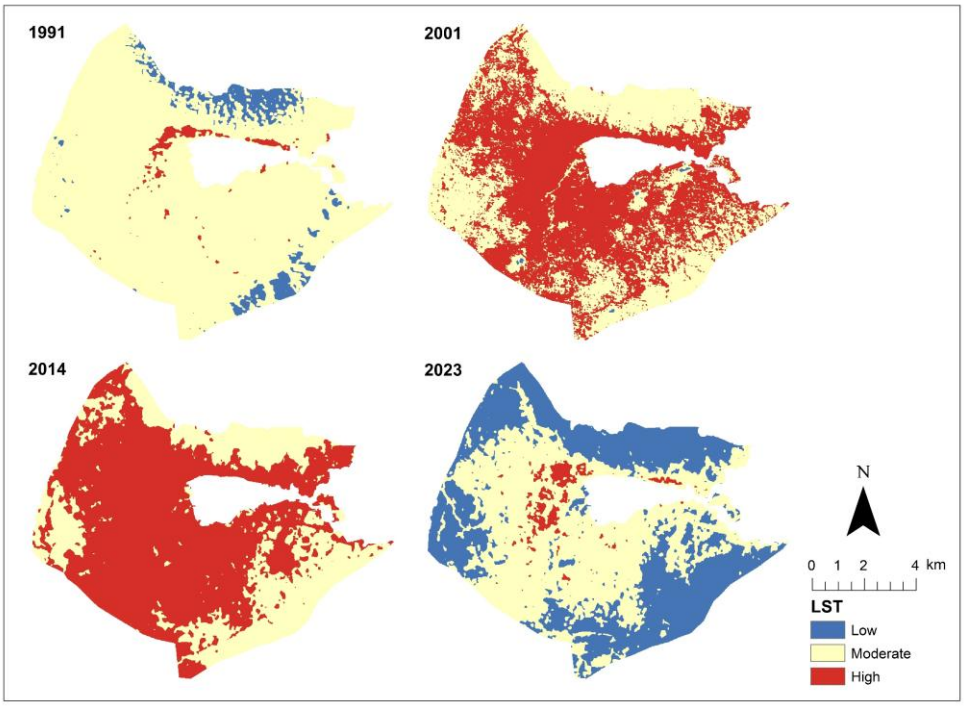
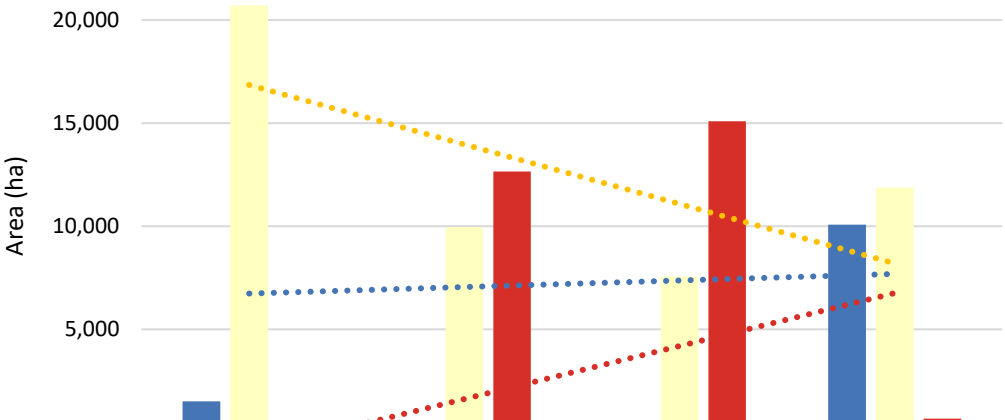


Figure 4. Distribution of Surface Temperature Index (LST) in Kendari City 1991-2023



	1991	2001	2014	2023
Low	1,513.71	33.30	-	10,085.58
Moderate	20,705.67	9,949.86	7,557.48	11,887.20
High	426.99	12,663.21	15,088.89	673.59

Figure 5. Distribution and Trend of LST Area in Kendari City 1991-2023

condition of the outermost part of an object on the land surface.

High-class LST distribution is concentrated in the central area, where dense infrastructure, including commercial buildings, offices, and housing, coincides with intense community activity (Figure 4). In contrast, the northern and southeastern suburban areas exhibit relatively lower temperatures, representing forest areas with high vegetation density that maintain a cooler climate. This spatial trend confirms that the urban core experiences significantly higher surface temperatures compared to its rural fringes.

The overall trend shows a decrease in low-class LST area, largely driven by the conversion of land into moderate and high-class thermal zones. While the low LST class reached 8,571.87 ha by 2023, the moderate class showed a continuous downward trend from 1991 to 2014 (Figure 5). Despite a slight recovery in cooler LST areas in 2023 (4,329.72 ha),

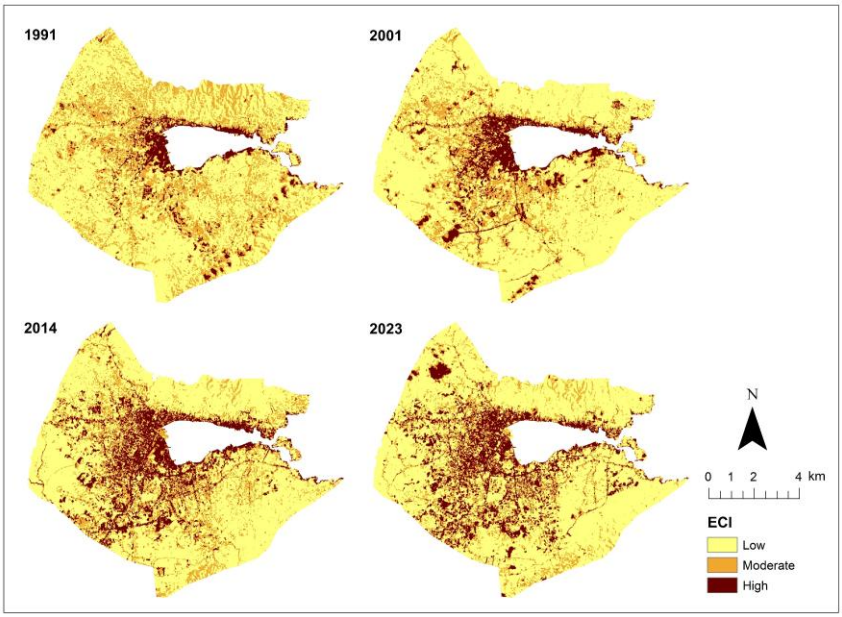


Figure 6. Distribution of Environmental Criticality Index (ECI) in Kendari City 1991-2023

the cumulative loss since 1991 remains substantial at 8,818.45 ha, suggesting that the decline in moderate LST areas is a direct result of their conversion into high-class thermal zones.

This phenomenon is indirectly related to the Urban Heat Island effect. While the World Health Organization (WHO) and the United Nations (UN-Habitat) consider air temperatures exceeding 35 °C as critical for human health [37]. The ECI in this study uses satellite-derived LST to measure surface thermal states rather than ambient air. In Kendari, LST values > 25 °C were predominantly recorded in highly built-up areas with low vegetation. Although these values sit below the 35 °C health threshold, they signify advanced surface warming and ecological degradation, as elevated surface temperatures reduce evapotranspiration and increase energy absorption, ultimately intensifying local heat stress.

Consequently, the spatial expansion of high-ECI zones serves as early evidence of environmental criticality, aligning with Montón Chiva & Quereda Sala [38] regarding the strong correlation between LST and city centers. While the densely populated central areas accumulate heat due to paved surfaces and low green cover, the suburbs remain cooler due to Kendari's unique geography between hills and the bay [35]. Vegetated areas stretching from the southwest to the west further mitigate temperatures by absorbing heat and reflecting solar radiation, creating a stark contrast to the urban core.

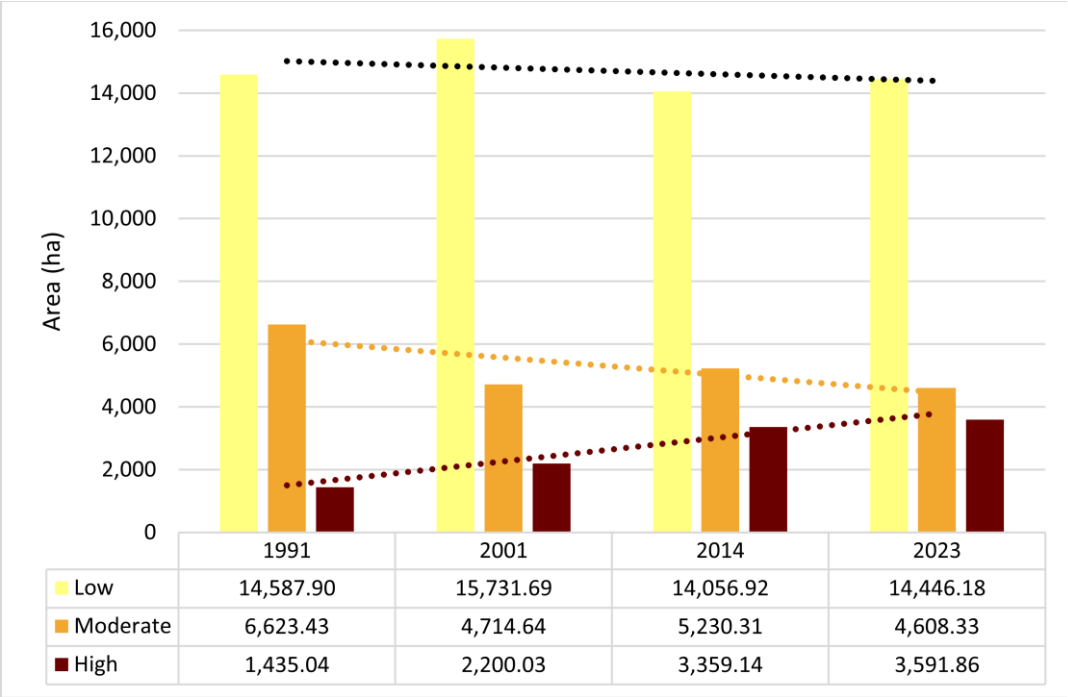
The patterns shown in Kendari City reflect an urbanization model where heat is concentrated in areas with minimal vegetation and dense buildings (Figure 4), as reported by Fikriyah et al. [39], that higher temperatures are in areas with denser buildings. Because concrete surfaces have a higher heat absorption capacity, the continued conversion of vegetated land to meet increasing spatial demands for land will inevitably alter spatial patterns and trigger a significant rise in urban air temperatures [40].

Environmental Criticality Index. The level of environmental criticality is obtained by calculating LST with NDVI.

While the low category dominates each analysis period, high-class ECI is concentrated in the city center along Kendari Bay (Figure 6), where the reduction of the vegetation canopy has led to higher surface temperatures compared to the cooler, high-vegetation suburbs. This categorization is driven by the stark contrast between the city center's high population density and minimal vegetation versus the suburbs' low density and abundant greenery.

Indirect validation confirms that areas with dense vegetation exhibit lower surface temperatures and ECI values, consistent with previous findings [22], [29], and despite the limitations of ground truth validation in this multitemporal study, the results maintain ecological and spatial consistency.

The low and moderate ECI classes show a fluctuating trend, whereas the high ECI class demonstrates a gradual and significant increase. In the low class, environmental criticality recovered from 1991 to 2001 by approximately 1,143.79 ha, decreased in 2014 by 1,674.77 ha, and recovered again in 2023 by 389.26 ha (Figure 7). A different, inversely proportional pattern is shown in the moderate class: critical land initially decreased to 1,908.79 ha in 2001, increased by 1/3 of the area to 515.67 ha in 2014, and then decreased again to 621.98 ha in 2023. Notably, the high ECI class continued to increase by



	1991	2001	2014	2023
Low	14,587.90	15,731.69	14,056.92	14,446.18
Moderate	6,623.43	4,714.64	5,230.31	4,608.33
High	1,435.04	2,200.03	3,359.14	3,591.86

Figure 7. Distribution and Trend of ECI Area in Kendari City 1991-2023

2,156.82 ha, with an average growth of 718.94 ha in each analysis period.

One area showing a very high ECI value is around Kota Lama (Figure 8a), an initial center of urban development where low albedo values lead to higher solar radiation absorption [41], coupled with dense settlements and minimal vegetation on the south side. In contrast, the area around the Halu Oleo University (Figure 8b) shows a unique ECI contraction. Despite being a development hub for the government center, the university maintains a green campus and city forest concept, resulting in high ECI values that are significantly lower than the surrounding new growth centers. Vegetation remains vital in reducing climate change effects [42], and the effectiveness of the NDVI algorithm in this study suggests its use as a key indicator for urban planning interventions.

To reduce ECI, several efforts are recommended: 1) integrating green space into urban spatial plans; 2) controlling built-up expansion in high-criticality

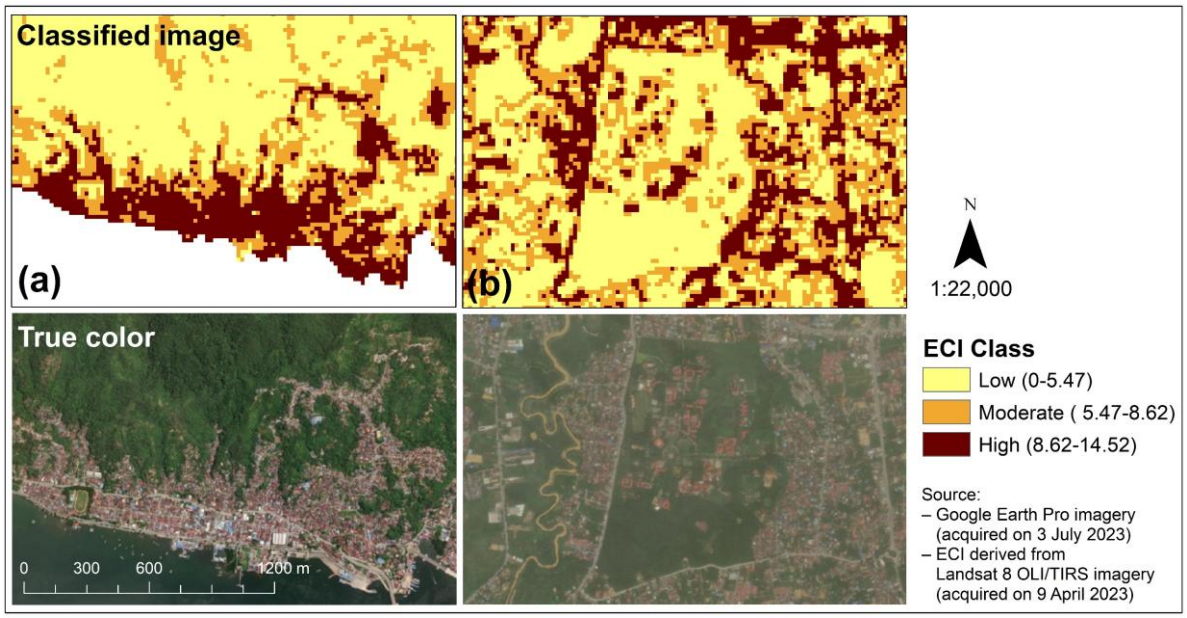


Figure 8. Areas with ECI class: a) Kota Lama, Kendari District, and b) Kampus Baru of Halu Oleo University, Andounohu District

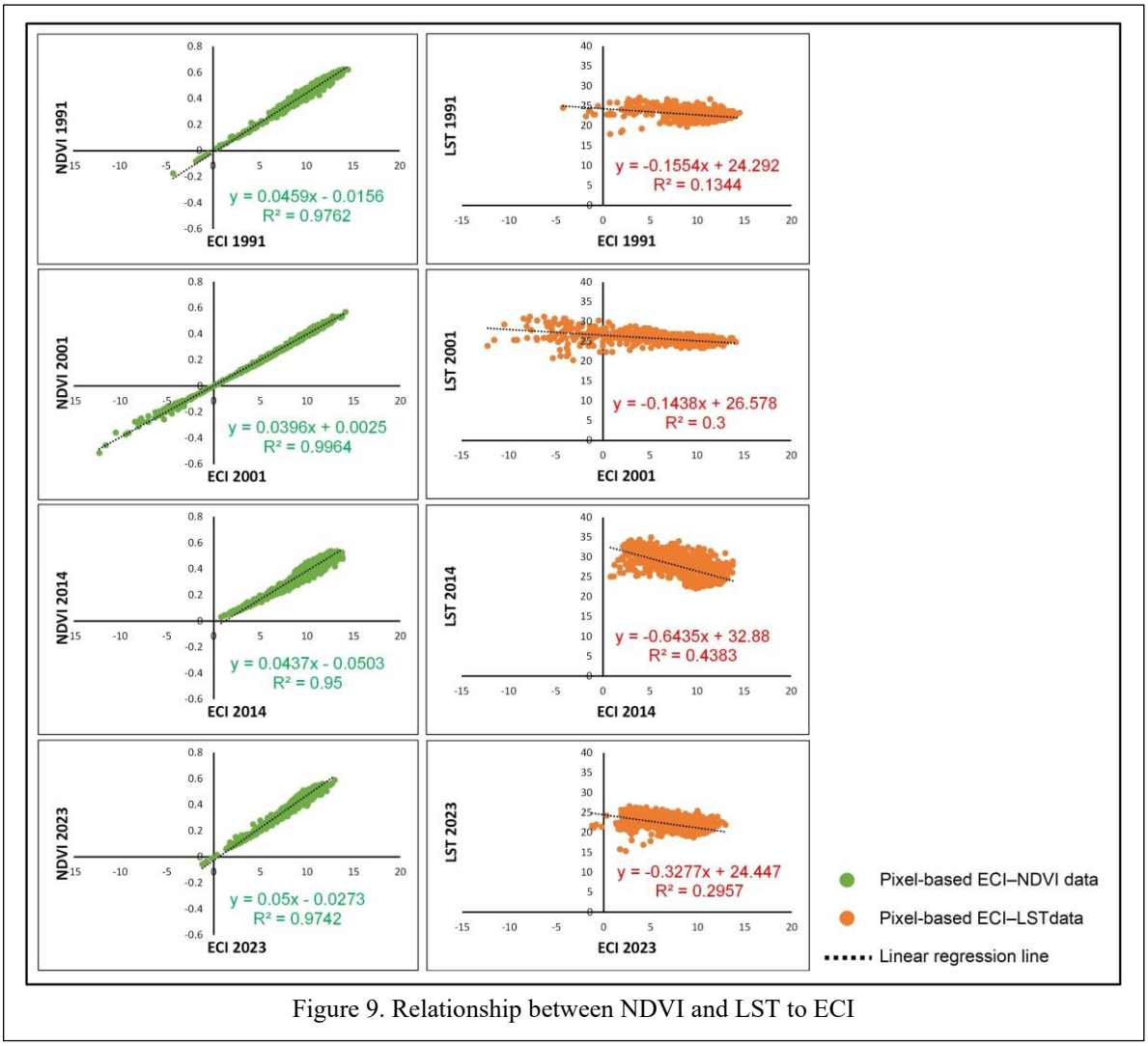


Figure 9. Relationship between NDVI and LST to ECI

zones; 3) implementing zoning in land-use planning; and 4) using ECI as a regular sustainability monitoring indicator, that strategies which have proven effective in previous studies [43], [44], [45].

While mapping the ECI through the integration of thermal and optical sensors provides an efficient spatial representation of land degradation, this study has inherent limitations due to its exclusive reliance on passive remote sensing, which primarily captures surface-level phenomena. The current multispectral data lacks the capacity to detect dynamic changes in water quality and groundwater levels, as optical sensors are constrained by their inability to penetrate the subsurface or aquifers. Furthermore, there is an absence of complex atmospheric parameters, such as aerosol concentrations or gaseous pollutants. Methodologically, the independent application of the Jenks Natural Breaks method poses challenges for radiometric standardization; since critical class thresholds are determined by the statistical distribution of pixel values for each specific acquisition, the analysis emphasizes relative spatial pattern shifts and expansion rather than inter-comparable absolute numerical values across different years.

Looking forward, the accuracy and depth of ECI studies could be significantly enhanced by adopting a Multi-Sensor Data Fusion approach. Integrating Synthetic Aperture Radar (SAR) data from Sentinel-1 would enable the monitoring of surface structures and soil moisture regardless of cloud cover, while data from the Sentinel-5P sensor could be incorporated to map air quality as an additional critical variable. To address the constraints in hydrological data, the use of derived indices such as the Modified Normalized Difference Water Index (MNDWI) or the utilization of gravity data from the GRACE satellite for regional groundwater storage estimation is highly recommended. Methodologically, future research should consider employing a Fixed Threshold based on long-term statistical normalization or Relative Radiometric Normalization (RRN) techniques. This would ensure that index fluctuations are quantitatively comparable over a consistent time-series, ultimately yielding a more precise environmental vulnerability prediction model for ecological disaster mitigation.

Relationship between Indicators. Further analysis was conducted to determine the correlation (R^2) of each ECI indicator to NDVI and LST. The coefficient values of NDVI to ECI from 1991 to 2023 were 0.9762, 0.9964, 0.95, and 0.9742, respectively (Figure 9). This indicates a very strong internal consistency since NDVI is a direct mathematical component of the index. This relationship reflects the structural coherence and spatial stability of the composite index, rather than suggesting that ECI statistically explains NDVI. Thus, statements regarding unexplained variance are not applicable here.

A positive linear association is observed between NDVI and ECI, with regression slopes of 0.0459, 0.0396, 0.0437, and 0.05 for 1991, 2001, 2014, and 2023, respectively (Figure 9), illustrating the index's sensitivity to vegetation greenness. For illustration, a small increase in NDVI (approx. 0.05 units), equivalent to 15–20 ha of additional vegetated surface in Kendari City, is associated with a decrease in ECI values of about 0.04–0.05 units. This highlights the stabilizing role of vegetation in moderating surface heat and reducing thermal stress, consistent with findings by Fadlin et al. [20], and other studies on urban environmental health [19].

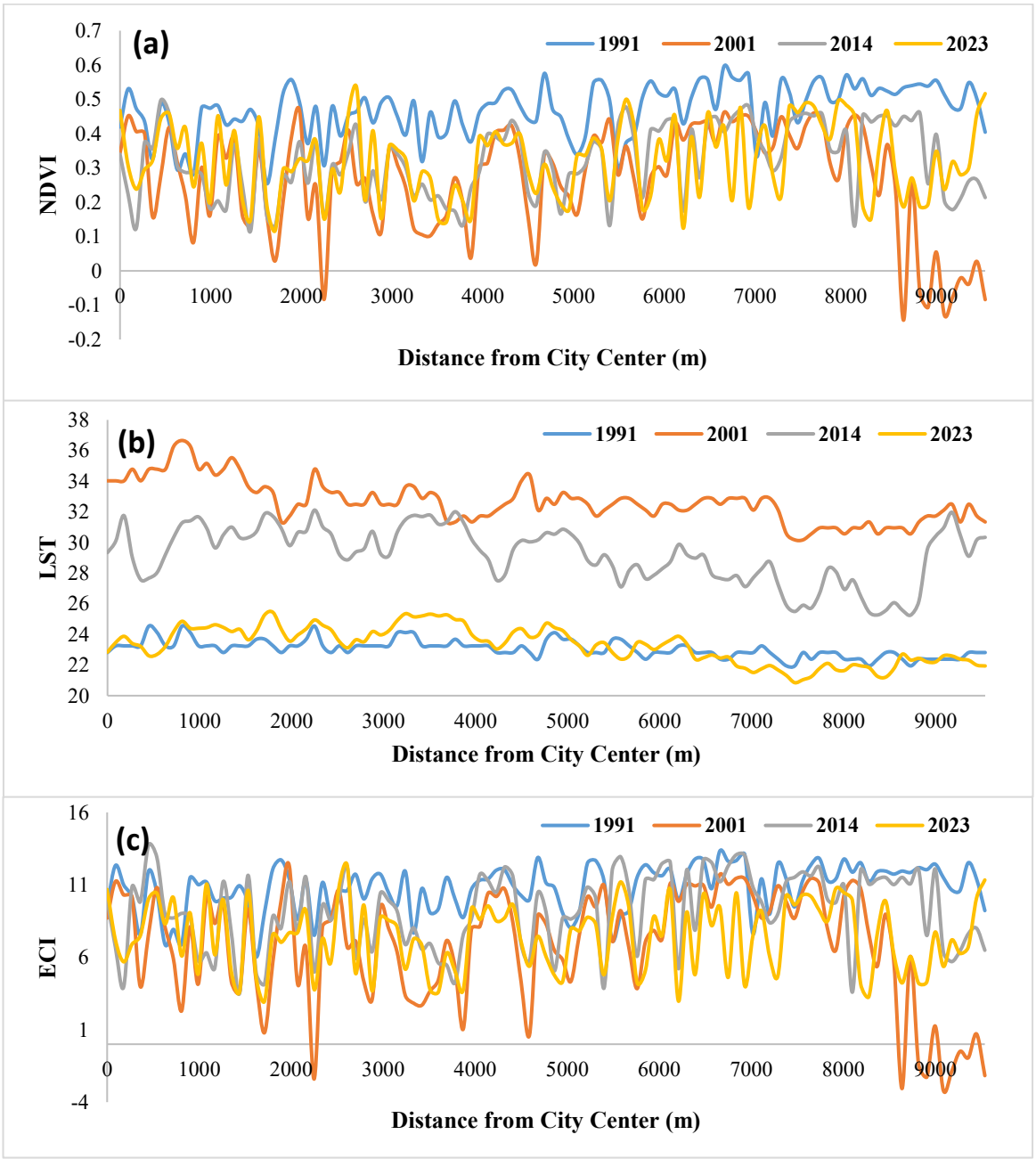


Figure 10. Urban-Rural line gradient pattern against a) NDVI, b) LST, and c) ECI

In contrast, the LST–ECI relationship exhibits lower, more dispersed R2 values of 0.1344, 0.3, 0.4383, and 0.2957 for the respective years (Figure 9), suggesting a moderate, indirect association rather than a dominant structural one. Lower surface temperatures (<35 °C) are consistently associated with lower ECI values, as reflected by negative regression coefficients of −0.1554 (1991), −0.1348 (2001), −0.6435 (2014), and −0.3277 (2023) (Figure 9). This indicates that while increasing temperatures are associated with higher criticality, the relationship is strongly mediated by land cover conditions and vegetation loss.

These results align with Aprilia & Mardia [46], confirming that the correlation between LST and ECI is weaker than the correlation between NDVI and ECI, due to vegetation's dominant role as a direct biophysical regulator. While vegetation enhances evapotranspiration and stabilizes microclimates, LST primarily serves as an indirect thermal response to land cover changes [47], [48]. Consequently, the influence of LST on ECI is largely mediated through urban land conversion [13], and supporting Pavlin et al. [49], in their assertion that vegetation management is essential for reducing environmental degradation and improving urban quality.

Urban-Rural Gradient. A gradient analysis was conducted along an urban-rural transect to observe the spatial patterns of NDVI, LST, and ECI. The resulting patterns show that NDVI values increase as one move toward rural areas (Figure 10a), whereas LST and ECI reach their peak in the urban core and decrease toward the suburbs (Figure 10b, 10c). This condition is closely related to land function where urban areas are dominated by buildings, resulting in lower NDVI values which indirectly elevate LST and ECI. These results are consistent with Ranagalage et al. [10], who reported a strong correlation between these indicators and urban physical expansion.

To quantify this spatial pattern, the transect was divided into buffer zones from the city center: 0–2 km (urban core), 2–5 km (suburban), and >5 km (rural zone). Data analysis reveals a clear transition with NDVI values increasing from approximately 0.21 in the urban core to 0.56 in suburban and 0.73 in rural areas. Simultaneously, LST decreased from 31.7 °C to 29.1 °C and 27.4 °C, respectively, while ECI values declined outward, indicating reduced environmental stress in the periphery. These findings suggest that if urban expansion continues beyond the 5 km buffer, current vegetated areas may transition into higher criticality zones.

These results align with Sustainable Development Goals (SDGs) 11 (Sustainable Cities and Communities) and 13 (Climate Action) by highlighting how vegetation cover and land conversion control improve urban livability and mitigate local climate impacts. Integrating green infrastructure is, therefore, critical for maintaining environmental resilience. Based on these findings, three key policy directions are proposed: 1) Prioritizing green infrastructure (urban forests and riparian vegetation) in expansion zones to buffer heat. 2) Stricter regulation of land conversion, particularly in the 2–5 km suburban transition zone where temperature thresholds are most sensitive. 3) Institutionalizing urban forest conservation and community reforestation to strengthen carbon sinks.

Overall, the spatial interaction among NDVI, LST, and ECI provides empirical evidence that vegetation-based interventions are vital for sustainable development. Integrating this evidence into Kendari's planning framework can transform the city into a model of green adaptation for Indonesia's rapidly developing secondary cities.

4. Conclusion

The analysis reveals that vegetation cover (NDVI) decreased by approximately 32.5%, while built-up and low-vegetation areas increased by more than 10%, particularly along the Kendari Bay corridor. Correspondingly, land surface temperature (LST) > 25 °C also expanded to 64.74%, and ECI values rose sharply in the urban core, indicating escalating environmental stress.

Spatially, ECI concentrations were highest within the 0–2 km urban core zone, decreasing toward suburban and rural areas along the urban–rural gradient. The strong correlation between NDVI and ECI ($R^2 > 0.95$) demonstrates that vegetation loss reflects increasing environmental criticality, while LST plays an indirect role by amplifying degradation through heat stress and reduced evapotranspiration.

This study is the first long-term ECI analysis of Kendari City based on multi-temporal satellite data, highlighting the novelty of applying an integrated NDVI–LST–ECI framework over more than three decades in a tropical coastal city with complex hilly topography. These findings provide new insights into how remote sensing can effectively capture both the spatial structure and temporal evolution of urban environmental criticality. The findings highlight the urgency of integrating green infrastructure, controlling land conversion near Kendari Bay, and enhancing urban vegetation buffers to mitigate surface warming and ecological decline.

From an environmental management perspective, maintaining or restoring vegetation can reduce land criticality and improve community resilience to urban heat and climate stress. Specifically, strategies such as the expanding of urban green belts, protecting coastal and riparian vegetation, and developing

interconnected green open spaces in high-criticality zones emerge as priority actions. Therefore, incorporating ECI monitoring into Kendari's spatial planning and sustainability evaluation system can guide more adaptive and evidence-based urban management.

Future research should integrate additional variables such as air quality, population density, and hydrological data, utilize higher-resolution or seasonal satellite imagery, and compare Kendari with other coastal urban systems to generalize the findings and strengthen the predictive capacity of urban sustainability assessments. In a broader context, long-term ECI monitoring has the potential to support Indonesian urban planning by providing a consistent, spatially explicit indicator for evaluating environmental carrying capacity and tracking the impacts of urban growth under climate and development pressures.

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